

HTTPv2 Face Recognition Device Backend Protocol Interface Document

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- Revised by Lai Jinjie

Revision History

Version	Revision Date	Description

HTTP Protocol Description

- Supports HTTP 1.1

- **GZIP Compression Technology**
- **Supports TLS 1.2, TLS 1.3**

HTTP API Interface

Call Flow

- The device first sends a heartbeat packet to the server, and the server responds with OK
- The device sends working parameters to the server
- After the server responds to the heartbeat keep-alive packet, the next request to initiate is determined based on the response content
- **Priority: syncParameter > Remote > deletePeople > addPeople**
- After processing the operations required by the server, check if there are accumulated unuploaded records
- If there are accumulated unuploaded records, push the records to the server
- The device must ensure that the interval between two heartbeat keep-alive packets does not exceed the set polling interval.
- i.e., if the heartbeat interval is 15 seconds, and the first heartbeat packet is sent at 10:00:00, and the server response includes deleting or adding people, the delete or add people interface should be called repeatedly, but ensure another heartbeat is sent at 10:00:15.
 - **Between two heartbeat intervals, process parameter synchronization, respond to remote operations, delete people, add people, and push records**

 image-20250421152526317

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About HTTP Connection Persistence

- The device and server should first establish a connection through the keep-alive packet. After the server responds successfully, the HTTP long connection should be maintained until all operations are completed before disconnecting. If there are no operations to perform, the connection can be disconnected immediately until the next keep-alive packet.
- The device only supports HTTP 1.1.
- A device should ensure that only one HTTP connection is maintained with the server to avoid concurrent multi-connections.

Compressed Requests

- This feature aims to reduce the size of the request body. When the device sends working parameters and attendance records, the JSON string is compressed using GZIP before sending.
- Designed for POST requests
- When compression is enabled, for requests with body of application/json, the request header will add Content-Encoding: gzip, and the body becomes binary data compressed using gzip.

```
POST /note/insertNoteFace HTTP/1.1
Accept: */*
Host: localhost:5000
Accept-Encoding: gzip, deflate, br
Connection: keep-alive
Content-Encoding: gzip
Content-Type: application/json
Content-Length: 842
```

****Compressed binary content****

- When the body is multipart, i.e., the format is multipart/form-data, the section containing JSON content will be compressed using gzip, and Content-Encoding: gzip will be added to the section header.

```
POST /note/insertNoteFace HTTP/1.1
Accept: */*
Host: localhost:5000
Accept-Encoding: gzip, deflate, br
Connection: keep-alive
Content-Type: multipart/form-data; boundary=-----
-506873351428002157394455
Content-Length: 17842
-----506873351428002157394455
Content-Disposition: form-data; name="SN"

FC-8380T12345678
-----506873351428002157394455
Content-Disposition: form-data; name="recordJson"
Content-Encoding: gzip
Content-Type: application/octet-stream

****Compressed binary content****
-----506873351428002157394455
Content-Disposition: form-data; name="pic"; filename="Postman_file.jpg"
Content-Type: image/jpeg

*****JPEG file binary content*****
-----506873351428002157394455--
```

Exception Handling

Network Exceptions

- 1. Unable to connect to the server at startup, sending upload parameter request fails
 - Record parameter upload status as not uploaded, communication enters sleep mode, waiting for the next interval
 - When the interval time arrives, send heartbeat packet, if the server does not respond, continue sleep waiting
 - After the heartbeat packet gets a response, check the parameter upload status; if not sent successfully, send once, then continue responding to subsequent operations requested by the server (refer to call flow)
- 2. Unable to connect to the server during device operation

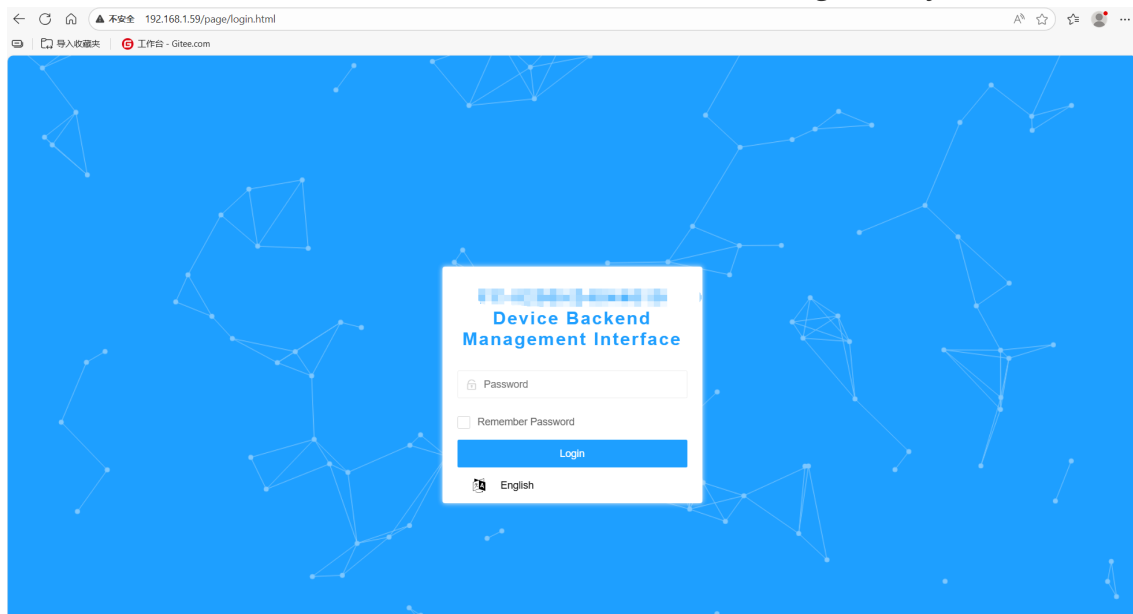
- Any request should include timeout and response retry -- recommended parameters: timeout 2 seconds, retry count 3 times
- After request failure, enter communication sleep mode, wait for the keep-alive packet sending interval
- For subsequent flow, refer to problem 1

Server Rejects Connection

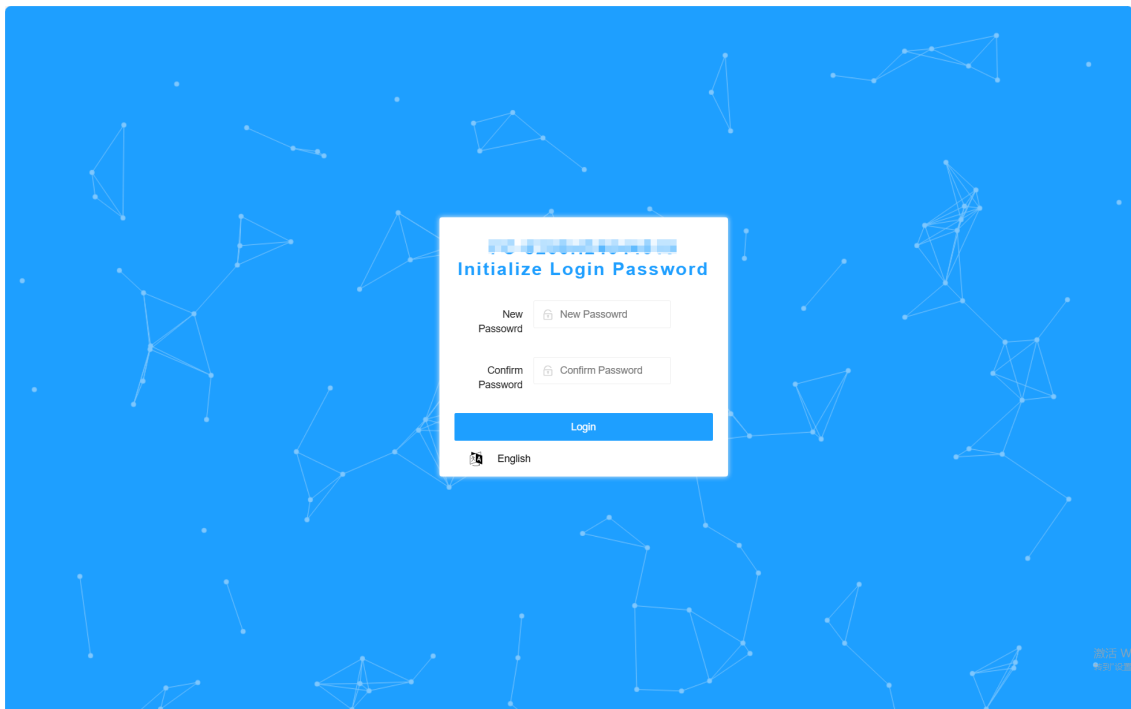
- After the device sends an attendance record request to the server API URL, in the response body returned by the server:
- If the Success field return value is 401/403, it indicates no permission. The device should enter request silence mode and wait for the next keep-alive cycle. After the keep-alive cycle, send a keep-alive first.
- When the Status in the header is 401/403/400 or other exceptions, also wait for the next keep-alive packet cycle, then send a keep-alive packet first to probe the server status.
- Only when the server returns HTTP status 200 and Success field is 1 can other API calls continue.

Server Configuration

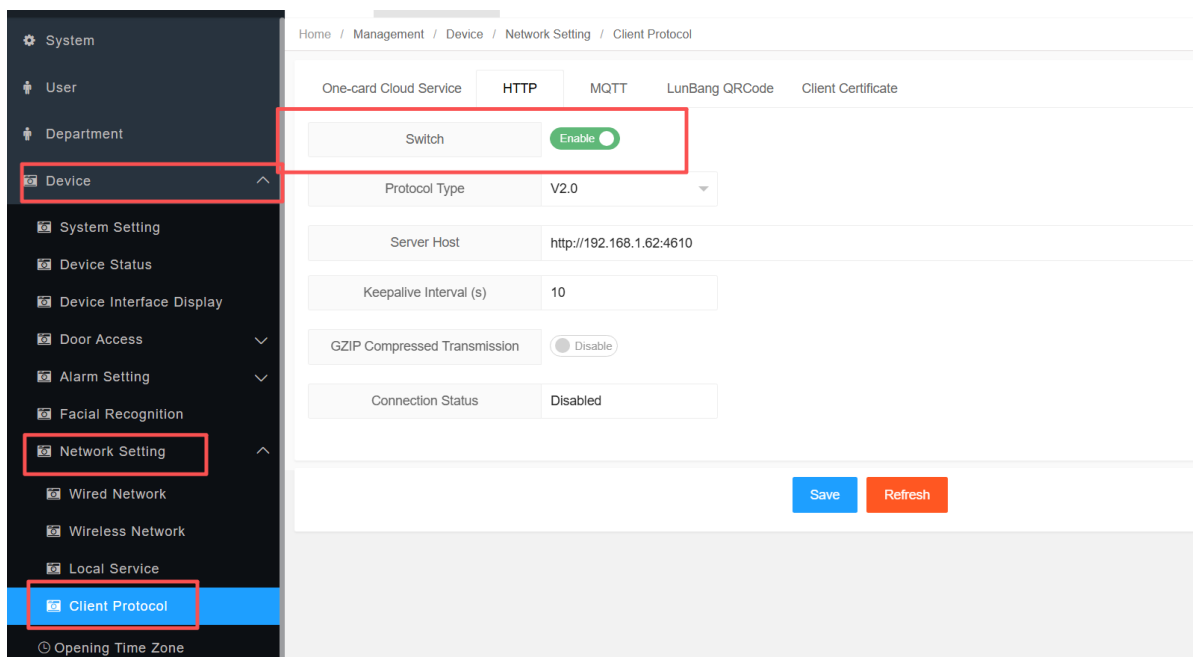
1. Use the device's IP address in a browser to access the device management system



2. If logging in for the first time, you need to first set a login password. The login password is the password for accessing the device menu interface.



3. Enter the "Network Settings" under the "Device" menu, select "Client Protocol", switch to the HTTP tab, and fill in the HTTP server address information on this page. The device will repeatedly attempt to access the server at the heartbeat interval.



Connection Keep-alive

API - Device Heartbeat Keep-Alive Packet

Brief Description:

- When the device is idle and the time since the last communication with the server exceeds the [Keep-alive Interval] seconds, a keep-alive packet is sent to confirm if there are any unfinished tasks.

Request URL:

- <http://Server IP:port/Device/Keepalive>

Request Method:

- POST

Content-Type

- application/json

Request Parameters

Field	Type	Description
SN	string	Device SN
RelayStatus	int	Relay physical status 0--indicates COM and NC normally closed 1--indicates COM and NO normally closed
KeepOpenStatus	int	Keep-open status 0--indicates normally closed 1--indicates keep-open
DoorSensorStatus	int	Door sensor status 0--indicates closed 1--indicates open
LockDoorStatus	int	Door lock status 0--indicates unlocked 1--indicates locked
AlarmStatus	string	Door alarm status Empty string means no alarm, otherwise there will be a specific alarm name fire--fire alarm blacklist--blacklist alarm anti--anti-disassembly alarm illegal--illegal verification password--duress alarm password openTimeout--door open timeout alarm doorSensor--door sensor alarm When there are multiple alarms, separate with commas: fire,blacklist

Request Parameter Example

```
{
  "SN": "FC-8200H12345678",
  "RelayStatus": 0,
  "KeepOpenStatus": 0,
  "DoorSensorStatus": 0,
  "LockDoorStatus": 0,
  "AlarmStatus": "",
}
```

Return Value Parameter Description

Parameter Name	Type	Required	Description
Success	int	Yes	1: success; 401/403 indicates connected to server but subsequent connections are rejected because the SN is not registered or installed
AddPeople	int	No	>0--indicates there are people to add to the device The device needs to initiate /People/DownloadPeopleList request upon receiving =0 or if this field is absent indicates no processing needed
DeletePeople	int	No	>0--indicates people need to be deleted from the device The device needs to initiate /People/DeletePeopleList request upon receiving
SyncParameter	int	No	1--indicates there are parameters to set on the device The device needs to initiate Device/DownloadWorkSetting request upon receiving
Remote	int	No	1--indicates there are remote operations to process The device needs to initiate /Device/RemoteCommand request upon receiving
UploadWorkParameter	int	No	1--indicates the server requires the device to upload device working parameters. The device needs to initiate Device/UploadWorkSetting request upon receiving

Return Value Example


```

{
  "Success":1,
  "AddPeople":1,
  "DeletePeople":1,
  "SyncParameter":1,
  "Remote":1,
  "UploadWorkParameter": 1
}

{
  "Success":1
}

```

- **Priority: UploadWorkParameter > Remote > SyncParameter > DeletePeople > AddPeople**

Return Error Codes

```

{
  "Success":401  //when returning 401, 400 or other non-0 values, it
indicates the device has encountered an issue on the platform. The device will no
longer send attendance records or device parameters to the server, but will still
periodically send keep-alive packets.
}
/*
when the device performs a test connection, the text prompted based on the return
value:
(1) 401 Prompt
① Chinese: 已连接服务器, 但是该设备未激活
② English: Connected to server, but device is not activated
(2) 400 Prompt
① Chinese: 已连接服务器, 但SN不符合平台规则
② English: Connected to server, but SN does not comply with platform rules
(3) 0 Prompt
① Chinese: 已连接服务器, 测试完毕
② English: Connected to server, testing completed

(4) Others: Prompt based on Message content
*/

```

Device Working Parameters

Reference file: [Device Working Parameters.md](#)

API - Upload Current Device Parameters

Brief Description:

- Sent once when the device starts up
- Uploaded again when device properties change

Request URL:

- <http://Server IP:port/Device/UploadWorkSetting>

Request Method:

- POST

Content-Type

- application/json

Request Parameters:

- Request Parameter Example

```
{
  "DevicesSN": "FC-8200H12345675",
  "FireAlarm": 0,
  "DoorLongOpenAlarm": 0,
  "DoorLongOpenTime": 0,
  "DoorSensorAlarm": 0,
  "DoorSensorAlarmTimegroup": {
    "week1": "",
    "week2": "",
    "week3": "",
    "week4": "",
    "week5": "",
    "week6": "",
    "week7": ""
  },
  "BlacklistAlarm": 1,
  "AntiDisassemblyAlarm": 1,
  "IllegalVerificationAlarm": 0,
  "IllegalVerificationAlarmLimit": 30,
  "UseUserCloseAlarm": 1,
  "PasswordAlarm": 0,
  "PasswordAlarm_Password": "",
  "PasswordAlarm_Mode": 1,
  "AlarmClocks": [
    {
      "Num": 1,
      "Clock": "00:00",
      "Times": 0
    },
    {
      "Num": 24,
      "Clock": "00:00",
      "Times": 0
    }
  ],
  "UseBodyTemperature": 1,
  "UseFahrenheitDisplay": 1,
  "TemperatureCompensate": 0,
  "TemperatureAlarmThreshold": 37.29999,
  "TemperatureDisplay": 1,
  "CardBytes": 3,
  "AccessType": 0,
  "WgFormat": 26,
  "WGContent": 1,
  "ReleaseTime": 3,
```

```
"FreeOpen": 0,
"OpenInterval": 2,
"OpenInterval_SaveRecord": 0,
"Relay": 0,
"ShortMessage": "",
"VerificationType": 1,
"OverdueRemind": 1,
"OverdueRemind_Day": 3,
"TimingOpen": 0,
"TimingOpen_mode": 1,
"TimingOpen_timegroup": {
    "week1": "",
    "week2": "",
    "week3": "",
    "week4": "",
    "week5": "",
    "week6": "",
    "week7": ""
},
"TimingLocked": 0,
"TimingLocked_timegroup": {
    "week1": "",
    "week2": "",
    "week3": "",
    "week4": "",
    "week5": "",
    "week6": "",
    "week7": ""
},
"VisitorRootPassword": "",
"MultiPerson": 1,
"UseElevator": 0,
"ElevatorPorts": [
    {
        "Num": 1,
        "RelayType": 2,
        "ReleaseTime": 2,
        "TimingOpen": {
            "Use": 0,
            "Open": 3,
            "Timegroup": {
                "week1": "",
                "week2": "",
                "week3": "",
                "week4": "",
                "week5": "",
                "week6": "",
                "week7": ""
            }
        }
    }
],
{
    "Num": 64,
    "RelayType": 2,
    "ReleaseTime": 2,
```

```
        "TimingOpen": {
            "Use": 0,
            "Open": 3,
            "Timegroup": {
                "week1": "",
                "week7": ""
            }
        }
    }
},
"FaceIR": 1,
"FaceIRThreshold": 5,
"FaceDistance": 3,
"FaceThreshold": 58,
"FPComparison": 80,
"FaceMask": 0,
"FaceMaskThreshold": 65,
"Holidays": [],
"Language": 1,
"SystemTime": 1718421632,
"UseNTP": 1,
"TimeZone": 8,
"Volume": 6,
"Voice": 1,
"ConnectPassword": "",
"UseWired": 1,
"WiredDHCP": 0,
"WiredIP": "192.168.1.103",
"WiredIPMask": "255.255.255.0",
"WiredGateway": "192.168.1.1",
"WiredDNS": "192.168.1.1",
"WiredMAC": "7E-87-16-C1-E6-51",
"UseWiFi": 0,
"WiFiDHCP": 0,
"WiFiIP": "192.168.1.150",
"WiFiIPMask": "255.255.255.0",
"WiFiGateway": "192.168.1.1",
"WiFiDNS": "192.168.1.1",
"WiFiMAC": "34-7D-E4-2D-63-3B",
"WiFiAPName": "",
"WiFiAPPassword": "",
"UseWebPage": 1,
"HTTPPort": 80,
"HTTPSPort": 443,
"WebPageUseSSL": 1,
"UseUDP": 1,
"UDPPort": 8101,
"UseTelnet": 1,
"TelnetPort": 23,
"UseRTSP": 1,
"RTSPPort": 554,
"RTSPUser": "admin",
"RTSPPassword": "12345678",
"UseTCPClient": 0,
"UseUDPClient": 0,
```

```
"ServerAddress": "47.92.31.75",
"ServerPort": 9003,
"KeepaliveTime": 30,
"UseHTTPClient": 0,
"HTTPClient_ProtocolType": 100,
"HTTPClient_ServerAddr": "http://192.168.1.100",
"HTTPClient_KeepaliveTime": 30,
"HTTPClient_UseGZIP": 0,
"UseMQTTClient": 0,
"UseMQTTSSL": 0,
"MQTTServerAddr": "192.168.1.100",
"MQTTPort": 0,
"MQTTLoginName": "",
"MQTTLoginPassword": "",
"MQTTPublishTopic": "",
"MQTTSubscribeTopic": "",
"MQTT_KeepaliveTime": 30,
"MQTT_UseGZIP": 0,
"UsewebsocketClient": 0,
"websocketClient_ProtocolType": 100,
"websocketClient_ServerAddr": "ws://192.168.1.100/ws",
"websocketClient_UseGZIP": 0,
"websocketClient_KeepaliveTime": 30,
"UseYZW": 0,
"YZWAddr": "http://192.168.1.10",
"RunDays": 0,
"FormatCount": 0,
"WatchDogCount": 7,
"BootTime": 1718418194,
"RelayStatus": 0,
"KeepOpenStatus": 0,
"DoorSensorStatus": 0,
"LockDoorStatus": 0,
"AlarmStatus": "\\\"",
"RecordAutoCycle": 0,
"SaveUnregistered": 1,
"SaveRecordPicture": 1,
"PeoplesStorageInfo": {
  "Person": {
    "Max": 20000,
    "Current": 1
  },
  "Face": {
    "Max": 20000,
    "Current": 1
  },
  "Card": {
    "Max": 20000,
    "Current": 0
  },
  "Fingerprint": {
    "Max": 0,
    "Current": 0
  },
  "Palmvein": {
```

```
        "Max": 10000,
        "Current": 0
    },
    "Pasword": {
        "Max": 20000,
        "Current": 0
    },
    "Admin": {
        "Max": 5,
        "Current": 0
    }
},
"RecordStorageInfo": {
    "VerifyRecord": {
        "Max": 1000000,
        "Current": 4
    },
    "DoorRecord": {
        "Max": 10000,
        "Current": 4
    },
    "SystemRecord": {
        "Max": 10000,
        "Current": 7
    },
    "RecordPhoto": {
        "Max": 10000,
        "Current": 4
    }
},
"DeviceSN": "FC-8190H24052799",
"DeviceName": "",
"FirmwareVerson": "8.46",
"FingerprintVerson": "-",
"FaceVerson": "6.01",
"Manufacturer": "",
"ManufacturerPhone": "",
"website": "",
"ProductionDate": "2024-06-15",
"OEMText": "",
"AutoRestart": 0,
"AutoRestartTime": "02:00",
"TimeGroups": [
    {
        "Num": 1,
        "week1": "00:00-23:59",
        "week2": "00:00-23:59",
        "week3": "00:00-23:59",
        "week4": "00:00-23:59",
        "week5": "00:00-23:59",
        "week6": "00:00-23:59",
        "week7": "00:00-23:59"
    },
    {
        "Num": 64,
```

```

        "week1": "",
        "week7": ""
    }
],
"DisplayBrightness": 6,
"MenuPassword": "0000",
"ShowIR": 0,
"ShowPersonPhoto": 1,
"PlayPersonName": 1,
"RecognitionButon": 0,
"UnregisteredWarn": 0,
"ShowPersonName": 1,
"FillLight": 2,
"FunctionList": {
    "BodyTemperature": true,
    "Fingerprint": false,
    "Palmvein": true,
    "Face": true,
    "QRCode": true,
    "FaceMask": true,
    "SafetyHelmet": false,
    "Lift": true,
    "AlarmClock": true,
    "ExcelFile": false,
    "ZipFile": false,
    "TimeGroup": 64,
    "WIFI": true,
    "HTTPClient_V1": true,
    "HTTPClient_V2": true,
    "MQTT": true,
    "YZW": true,
    "Websocket_V1": true,
    "Websocket_V2": true
}
}

```

Return Parameter Description

Parameter Name	Type	Required	Description
Success	int	Yes	1 indicates operation successful; != 1 requires error description in Message
Message	string	No	Error information description

Return Value Example

```

{
    "Success":1
}

```

API - Device Actively Pulls Working Parameters

Brief Description:

- After the device sends a heartbeat packet, if the SyncParameter field in the server response is 1, the device will request this interface.

Request URL:

- <http://Server IP:port/Device/DownloadWorkSetting>

Request Method:

- POST

Content-Type

- application/json

Request Parameter Description

Parameter Name	Type	Required	Description
SN	string	Yes	Device ID

Request Parameter Example

```
{
  "SN": "FC-8200H12345678"
}
```

Return Parameter Description

Field	Type	Required	Description
Success	int	Yes	1--indicates success Others are error codes, need to indicate error content in Message
Message	string	No	Error information description, see Chapter 3 for details
All modifiable parameters			

Return Value Example

```
{
  "Success": 1,
  "DevicesSN": "FC-8200H12345675",
  "FireAlarm": 0,
  "DoorLongOpenAlarm": 0,
  "DoorLongOpenTime": 0,
  "DoorSensorAlarm": 0,
  "DoorSensorAlarmTimegroup": {
    "week1": "",
    "week2": "",
    "week3": "",
    "week4": "",
    "week5": ""
  }
}
```



```
        "week6": "",
        "week7": ""
    },
    "BlacklistAlarm": 1,
    "AntiDisassemblyAlarm": 1,
    "IllegalVerificationAlarm": 0,
    "IllegalVerificationAlarmLimit": 30,
    "UseUserCloseAlarm": 1,
    "PasswordAlarm": 0,
    "PasswordAlarm_Password": "",
    "PasswordAlarm_Mode": 1,
    "AlarmClocks": [
        {
            "Num": 1,
            "Clock": "00:00",
            "Times": 0
        },
        {
            "Num": 24,
            "Clock": "00:00",
            "Times": 0
        }
    ],
    "UseBodyTemperature": 1,
    "UseFahrenheitDisplay": 1,
    "TemperatureCompensate": 0,
    "TemperatureAlarmThreshold": 37.29999,
    "TemperatureDisplay": 1,
    "CardBytes": 3,
    "AccessType": 0,
    "WgFormat": 26,
    "WGContent": 1,
    "ReleaseTime": 3,
    "FreeOpen": 0,
    "OpenInterval": 2,
    "OpenInterval_SaveRecord": 0,
    "Relay": 0,
    "ShortMessage": "",
    "VerificationType": 1,
    "OverdueRemind": 1,
    "OverdueRemind_Day": 3,
    "TimingOpen": 0,
    "TimingOpen_mode": 1,
    "TimingOpen_timegroup": {
        "week1": "",
        "week2": "",
        "week3": "",
        "week4": "",
        "week5": "",
        "week6": "",
        "week7": ""
    },
    "TimingLocked": 0,
    "TimingLocked_timegroup": {
        "week1": "",
```

```
        "week2": "",
        "week3": "",
        "week4": "",
        "week5": "",
        "week6": "",
        "week7": ""
    },
    "VisitorRootPassword": "",
    "MultiPerson": 1,
    "UseElevator": 0,
    "ElevatorPorts": [
        {
            "Num": 1,
            "RelayType": 2,
            "ReleaseTime": 2,
            "TimingOpen": {
                "Use": 0,
                "Open": 3,
                "Timegroup": {
                    "week1": "",
                    "week2": "",
                    "week3": "",
                    "week4": "",
                    "week5": "",
                    "week6": "",
                    "week7": ""
                }
            }
        }
    ],
    {
        "Num": 64,
        "RelayType": 2,
        "ReleaseTime": 2,
        "TimingOpen": {
            "Use": 0,
            "Open": 3,
            "Timegroup": {
                "week1": "",
                "week7": ""
            }
        }
    }
],
    "FaceIR": 1,
    "FaceIRThreshold": 5,
    "FaceDistance": 3,
    "FaceThreshold": 58,
    "FPComparison": 80,
    "FaceMask": 0,
    "FaceMaskThreshold": 65,
    "Holidays": [],
    "Language": 1,
    "SystemTime": 1718421632,
    "UseNTP": 1,
    "TimeZone": 8,
```

```
"Volume": 6,
"Voice": 1,
"ConnectPassword": "",
"UseWired": 1,
"WiredDHCP": 0,
"WiredIP": "192.168.1.103",
"WiredIPMask": "255.255.255.0",
"WiredGateway": "192.168.1.1",
"WiredDNS": "192.168.1.1",
"WiredMAC": "7E-87-16-C1-E6-51",
"UseWiFi": 0,
"WiFiDHCP": 0,
"WiFiIP": "192.168.1.150",
"WiFiIPMask": "255.255.255.0",
"WiFiGateway": "192.168.1.1",
"WiFiDNS": "192.168.1.1",
"WiFiMAC": "34-7D-E4-2D-63-3B",
"WiFiAPName": "",
"WiFiAPPassword": "",
"UseWebPage": 1,
"HTTPPort": 80,
"HTTPSPort": 443,
"WebPageUseSSL": 1,
"UseUDP": 1,
"UDPPort": 8101,
"UseTelnet": 1,
"TelnetPort": 23,
"UseRTSP": 1,
"RTSPPort": 554,
"RTSPUser": "admin",
"RTSPPassword": "12345678",
"UseTCPClient": 0,
"UseUDPClient": 0,
"ServerAddress": "47.92.31.75",
"ServerPort": 9003,
"KeepaliveTime": 30,
"UseHTTPClient": 0,
"HTTPClient_ProtocolType": 100,
"HTTPClient_ServerAddr": "http://192.168.1.100",
"HTTPClient_KeepaliveTime": 30,
"HTTPClient_UseGZIP": 0,
"UseMQTTClient": 0,
"UseMQTTSSL": 0,
"MQTTServerAddr": "192.168.1.100",
"MQTTPort": 0,
"MQTTLoginName": "",
"MQTTLoginPassword": "",
"MQTTPublishTopic": "",
"MQTTSubscribeTopic": "",
"MQTT_KeepaliveTime": 30,
"MQTT_UseGZIP": 0,
"UseWebSocketClient": 0,
"WebSocketClient_ProtocolType": 100,
"WebSocketClient_ServerAddr": "ws://192.168.1.100/ws",
"WebSocketClient_UseGZIP": 0,
```

```

"WebSocketClient_KeepaliveTime": 30,
"UseYZW": 0,
"YZWAddr": "http://192.168.1.10",
"RecordAutoCycle": 0,
"SaveUnregistered": 1,
"SaveRecordPicture": 1,
"DeviceName": "",
"Manufacturer": "",
"ManufacturerPhone": "",
"Website": "",
"ProductionDate": "2024-06-15",
"OEMText": "",
"AutoRestart": 0,
"AutoRestartTime": "02:00",
"TimeGroups": [
  {
    "Num": 1,
    "Week1": "00:00-23:59",
    "Week2": "00:00-23:59",
    "Week3": "00:00-23:59",
    "Week4": "00:00-23:59",
    "Week5": "00:00-23:59",
    "Week6": "00:00-23:59",
    "Week7": "00:00-23:59"
  },
  {
    "Num": 64,
    "Week1": "",
    "Week7": ""
  }
],
"DisplayBrightness": 6,
"MenuPassword": "0000",
"ShowIR": 0,
"ShowPersonPhoto": 1,
"PlayPersonName": 1,
"RecognitionButon": 0,
"UnregisteredWarn": 0,
"ShowPersonName": 1,
"FillLight": 2
}

```

Remote Control

API - Remote Operation Commands

Brief Description:

- When the Remote value in the heartbeat packet return value is 1, the device immediately requests this interface to obtain remote operation commands.

Request URL:

- <http://Server IP:port/Device/RemoteCommand>

Request Method:

- POST

Content-Type

- application/json

Request Parameters:

Field	Type	Length	Required	Description
SN	string	30	Yes	Device ID

Request Parameter Example

```
{
  "SN": "FC-8200H12345678"
}
```

Return Value Parameter Description

Parameter Name	Type	Required	Description
Success	int	Yes	1--indicates success Others are error codes, need to indicate error content in Message; 0--indicates failure
Message	string	No	Error information description, see Chapter 3 for details
Restart	int	No	Remote restart `0: no restart, 1: restart
Recover	int	No	Factory reset ``0: normal, 1: restore factory settings
Opendoor	int	No	Remote door open command `0: no action, 1--activate relay; 2--keep door open; 3--close door (cancel keep-open); `4--lock door; 5--unlock door
Closealarm	int	No	Close alarm command ``0: no action, 1: close all active alarms and record
RepostRecord	int	No	Re-upload records ``0: no action, 1: re-mark all uploaded records as not uploaded and re-transmit
PushAllPeople	int	No	Request to upload all stored person lists to server Device calls API [/People/PushPeople] to send person list 0--no action needed, 1--need to upload all people.

Parameter Name	Type	Required	Description
QueryPeople	[uint]	No	Request to upload specified user ID people to server Device calls API [/People/PushPeople] to send person list Type is array
ClearRecord	int	No	Delete all records; 0--no action, 1--delete all records
RegisterIdentifyTicket	object	No	Register credential type on device See RegisterIdentifyTicket Command Register Credential Object Structure for object structure After registration, call API /Device/RegisterIdentifyTicketResult to upload result
PushSoftware	object	No	Push software firmware upgrade package download URL. After receiving the URL, the device downloads the upgrade package, and automatically updates the firmware after successful verification. See PushSoftware Command Firmware File Object Structure for object structure
PushSystemFile	[object]	No	Push system file update information, Data type is array See PushSystemFile Command System File Object Structure for each array element object structure
Snapshot	int	No	Get device camera snapshot; 0--no action, 1--take camera snapshot and upload Upload snapshot by calling API /Device/RegisterIdentifyTicketResult
OpenElevatorPort	string	No	Remotely drive an elevator port once Comma-separated string: 1,2,3,4,5
KeepOpenElevatorPort	string	No	Put elevator port into keep-open state Comma-separated string: 1,2,3,4,5
CloseElevatorPort	string	No	Exit elevator port from keep-open state Comma-separated string: 1,2,3,4,5
LockElevatorPort	string	No	Put elevator port into locked state, users cannot drive when locked Comma-separated string: 1,2,3,4,5
UnlockElevatorPort	string	No	Exit elevator port from locked state Comma-separated string: 1,2,3,4,5
UploadRecordRange	object	No	Upload records within a specified time range to server

RegisterIdentifyTicket Command - Register Credential Object Structure

Field	Type	Required	Description
RegisterType	uint	Yes	Credential type to register See RegisterType table
UserID	uint	No	User ID for registering credential on device
RegisterIndex	uint	No	Credential index number for registration For fingerprint, index number is 0-9 For palm vein, index number is 1-2

RegisterType

Value	Description
1	Register Card
2	Register Password
3	Register Fingerprint
4	Register Face
5	Register Palm Vein

PushSoftware Command - Firmware File Object Structure

Field	Type	Length	Required	Description
SoftwareURL	string	1024	Yes	File URL address
SoftwareMD5	string	64	Yes	File MD5 hex encoded format, uppercase
SoftwareVer	string		Yes	File firmware version number

PushSystemFile Command - System File Object Structure

Field	Type	Length	Required	Description
FileURL	string	1024	Yes	File URL address
FileMD5	string	64	Yes	File MD5 hex encoded format, uppercase
FileType	int	1	Yes	File type See FileType for file type definitions
FileIndex	int	1	Yes	File index
IsDelete	int	1	Yes	Whether it indicates file deletion

FileType

Value	Description
1	Standby image, up to 8 images Supported index range 1-8
2	Boot image, only one image

UploadRecordRange Command - Time Range Object Structure

Field	Type	Length	Required	Description
BeginTime	long		Yes	Start timestamp of time range
EndTime	long		Yes	End timestamp of time range

Return Value Examples

```
//Remote door open
{
  "Success":1,
  "Opendoor":1
}

//Query specified people
{
  "Success":1,
  "QueryPeople":[1,2,3,4,5,6]
}

//Read records for a specific time range
{
  "UploadRecordRange" : {
    "BeginTime" : 123456, //Record start time
    "EndTime" : 123456,   //Record end time
  }
}

//Register credential -- register face for user ID 1
{
  "Success":1,
  "RegisterIdentifyTicket": 4,
  "RegisterIdentifyUserID": 1
}
```


API - Feedback Person Register Credential Result

Brief Description:

- After receiving the remote command to register a person's credential, the device enters the credential registration interface and waits for the operation result. After registration is complete, call this API to return the person credential registration result.
- And call the Push Person API to push the latest person information after registration.
- Uses form upload method, Content-Type: multipart/form-data, the returned result may contain images.

Request URL:

- http://Server_IP:port/Device/RegisterIdentifyTicketResult

Request Method:

- POST

Content-Type

- multipart/form-data

form-data Fields

Field	Type	Length	Required	Description
ResultJson	string		Yes	Registration result JSON string
Photo	file		No	Returns photo when person has a photo

Resultjson Structure:

Field	Type	Length	Required	Description
SN	string	30	Yes	Device ID
UserID	uint		Yes	User ID
Result	int		Yes	Registration credential result 1. Registration successful; 2. User cancelled operation; 3--Registration information duplicate 4--Fingerprint not supported 5--Palm vein not supported 6--Device busy
RepetitionUserID	uint		No	When registered credential is duplicate, shows the duplicate user ID
UserDetail	object		No	When registration is successful, returns the latest person details. If there is a person photo the person photo will be returned

Request Parameter Examples

```
//Duplicate registration return
{
  "SN":"FC-8200H12345678",
  "UserID": 1,
  "Result": 2,
  "RepetitionUserID": 2
}

//Successful registration return
{
  "SN":"FC-8200H12345678",
  "UserID": 1,
  "Result": 1,
  "UserDetail": {
    "UserID": "3",
    "Name": "888888",
    "Job": "Development",
    "Department": "Sales Department",
    "IdentityCard": "",
    "Attachment": "",
    "Photo": "http://abc.com/photo/1.jpg",
    "PhotoMD5": "613D870CA99EDF074BEE4387BAB09070",
    "PhotoLen": 55020,
    "Password": "2222",
    "CardNum": "6666",
    "AccessType": 0,
    "ExpirationDate": 0,
    "OpenTimes": 65535,
    "KeepOpen": 1,
    "Timegroup": 6,
    "Holidays": "1,3,9,10,11,17,21,25,27,30",
    "Elevators": "1,2,3,4,5",
    "FaceFeature": "http://abc.com/face/1.dat",
    "FaceFeatureMD5": "613D870CA99EDF074BEE4387BAB09070",
    "Fingerprints": [
      {
        Num: 1,
        Data: "abcdefgrtygnfhgfhjk ... jhkghjkgj==",
        MD5: "abcdefg"
      }
    ],
    "Palmveins": [
      {
        Num: 1,
        Data: "abcdefgrtygnfhgfhjk ... jhkghjkgj==",
        MD5: "abcdefg"
      }
    ]
  }
}
```

Return Value Parameter Description

Parameter Name	Type	Required	Description
Success	int	Yes	1--indicates success Others are error codes, need to indicate error content in Message; 0--indicates failure
Message	string	No	Error information description, see Chapter 3 for details

Return Value Example

```
{  
  "Success": 1  
}
```

API - Upload Device Camera Snapshot

Brief Description:

- After receiving the remote command to get a camera snapshot, the device immediately takes a snapshot of the camera and calls this interface to upload.
- Uses form upload method, Content-Type: multipart/form-data, the returned result contains an image.

Request URL:

- http://Server_IP:port/Device/UploadSnapshot

Request Method:

- POST

Content-Type

- multipart/form-data

form-data Fields

Field	Type	Length	Required	Description
SN	string		Yes	Device SN
Photo	file		No	Device camera snapshot

Return Value Parameter Description

Parameter Name	Type	Required	Description
Success	int	Yes	1--indicates success Others are error codes, need to indicate error content in Message; 0--indicates failure
Message	string	No	Error information description, see Chapter 3 for details

Return Value Example

```
{  
  "Success": 1  
}
```

People Management

API - Pull People Authorization Information

Brief Description:

- When the heartbeat keep-alive server return value contains a field indicating people need to be pulled and the value is 1, the device immediately initiates this request and loops the request.
- Stop request conditions: 1. Server returns Success:1 but the people list is empty; 2. Success==0

Request URL:

- http://Server_IP:port/People/DownloadPeopleList

Request Method:

- POST

Content-Type

- application/json

Request Parameters

Field	Type	Length	Required	Description
SN	string	30	Yes	Device ID

Field	Type	Length	Required	Description
Limit	int	10	Yes	Limit on number of people returned per request (max 1000) ``Device sets this value based on its own processing capacity

Request Parameter Example

```
{
  "SN": "FC-8200H12345678",
  "Limit": 100
}
```

Return Value Parameter Description

Parameter Name	Type	Required	Description
Success	int	Yes	1--indicates success Others are error codes, need to indicate error content in Message; 0--indicates failure
Message	string	No	Error information description, see Chapter 3 for details
PeopleCount	int	Yes	Number of people returned this time
PeopleList	array	Yes	Array containing person permission information structure

Peoplejson Person Data Format

Parameter Name	Type	Required	Description
UserID	string	Yes	User ID (number, max value 4294967295, type UINT32)
Code	string	No	Person code (characters <64)
Name	string	No	Person name (characters <64)
Job	string	No	Job title (characters <64)
Department	string	No	Department (characters <64)
IdentityCard	string	No	ID card number (can be empty) (characters <64)
Attachment	string	No	Other person information (characters <200)

Parameter Name	Type	Required	Description
Photo	string		Photo information, http or https prefix indicates URL address, otherwise indicates base64
PhotoMD5	string	No	Photo MD5 HEX string format (characters=32, required if photo information exists)
PhotoLen	int	No	Photo length, max supported 400KB image (required if photo information exists)
Password	string	No	Password, digits only, length: (0 / 4-8)
CardNum	string	No	Card number (number, max value 18446744073709551615, type UINT64)
QRCode	string	No	Person QR code information (characters <128)
AccessType	int	No	Role 0--normal person; 1--admin; 2--blacklist
ExpirationDate	uint32	No	Permission expiration date Unix timestamp in seconds, 0 indicates unlimited Max represents December 31, 2099
OpenTimes	int	No	Door open count 0-65535; 65535--indicates unlimited, 0--indicates access forbidden
KeepOpen	int	No	Whether it's a keep-open card, 1--yes; 0--no
Timegroup	int	No	Access time group 1-64; 0--indicates restricted
Holidays	string	No	Holiday restriction Comma-separated: 1,2,3,4,5
Elevators	string	No	Elevator port permission array Comma-separated: 1,2,3,4,5
FaceFeature	string	No	Face feature code, http or https prefix indicates URL address, otherwise indicates base64 Feature code downloaded via file, downloaded file content is feature code binary
FaceFeatureMD5	string	No	Face feature code MD5 value, HEX string format
Fingerprints	[]	No	Fingerprint object
Palmveins	[]	No	Palm vein object

Characters refer to bytes. One Chinese character occupies 3-4 bytes (UTF-8 encoding), one English character occupies 1 byte.

Holidays

- Empty string or absence of this field indicates no holiday restrictions
- Specific restricted holiday numbers, comma-separated: 1,2,3,4,5
- - indicates restricted by all holidays

- **Fingerprint Object**

No.	Type	Required	Description
Num	int	Yes	Fingerprint index number
Data	string	Yes	Fingerprint feature code, http or https prefix indicates URL address, otherwise indicates base64 Feature code downloaded via file, downloaded file content is feature code binary
MD5	string	No	Feature code MD5 value, HEX string format

```
[ //Structure example
  {
    Num: 1,
    Data: "http://abc.com/fp/1.dat",
    MD5: "abcdefg"
  },
  {
    Num: 2,
    Data: "abcdefgrtygnfhgfjkh ... jhkgjhkgj==",
    MD5: "abcdefg"
  }
]
```

- **Palm Vein Object**

No.	Type	Required	Description
Num	int	Yes	Palm vein index number
Data	string	Yes	Palm vein feature code, http or https prefix indicates URL address, otherwise indicates base64 Feature code downloaded via file, downloaded file content is feature code binary
MD5	string	No	Feature code MD5 value, HEX string format

```
[ //Structure example
  {
    Num: 1,
    Data: "http://abc.com/palm/1.dat",
    MD5: "abcdefg"
  },
  {
    Num: 2,
    Data: "abcdefgrtygnfhgfjkh ... jhkgghjkghj==",
    MD5: "abcdefg"
  }
]
```

- Elevator Permissions

```
//Indicates this person has elevator permission for floors 1-5
[
  1,2,3,4,5
]
//Indicates this person has no elevator permission
[
]
//Indicates this person only has elevator permission for floor 10
[
  10
]
```

Return Parameter Example

```
{
  "Success": 1,
  "Count": 1,
  "PeopleList": [
    {
      "UserID": "3",
      "Name": "888888",
      "Job": "Development",
      "Department": "Sales Department",
      "IdentityCard": "",
      "Attachment": "",
      "Photo": "http://abc.com/photo/1.jpg",
      "PhotoMD5": "613D870CA99EDF074BEE4387BAB09070",
      "PhotoLen": 55020,
      "Password": "2222",
      "CardNum": "6666",
      "AccessType": 0,
      "ExpirationDate": 0,
      "OpenTimes": 65535,
      "KeepOpen": 1,
      "Timegroup": 6,
      "Holidays": "1,3,9,10,11,17,21,25,27,30",
    }
  ]
}
```



```
    "Elevators": "1,2,3,4,5",
    "FaceFeature": "http://abc.com/face/1.dat",
    "FaceFeatureMD5": "613D870CA99EDF074BEE4387BAB09070",
    "Fingerprints": [
      {
        Num: 1,
        Data: "http://abc.com/fp/1.dat",
        MD5: "abcdefg"
      }
    ],
    "Palmveins": [
      {
        Num: 1,
        Data: "http://abc.com/palm/1.dat",
        MD5: "abcdefg"
      }
    ]
  },
  {
    "UserID": "444",
    "Name": "444",
    ...
    "Fingerprints": [],
    "Palmveins": []
  }
]
```

API - Feedback Pull People Storage Result

Brief Description:

- After pulling a batch of people from the server and importing them, call this interface to return the import result.

Request URL:

- <http://Server IP:port/People/DownloadPeopleListResult>

Request Method:

- POST

Content-Type

- application/json

Request Parameters:

Field	Type	Length	Required	Description
SN	string	30	Yes	Device ID
SuccessCount	int	20	Yes	Number of successful imports
FailCount	int	20	Yes	Number of failed imports

Field	Type	Length	Required	Description
FailList	[]		Yes	Import failure reason codes (see error codes for specific descriptions)

Failure information structure in FailList array

Parameter Name	Type	Required	Description
UserID	string	Yes	User ID (number, max value 4294967295, type UINT32)
ErrorCode	int	Yes	Error code
RepeatID	string	Yes	Duplicate user ID, this field indicates who this person is duplicated with
ErrMsg	string	No	Error description

Request Parameter Example

```
{
  "SN": "FC-8200H12345678",
  "SuccessCount": 16,
  "FailCount": 1,
  "FailCount": [
    {
      "UserID": xxx,
      "ErrorCode": 20,
      "RepeatID": 123,
      "ErrMsg" : ""
    }, {
      "UserID": "xxx",
      "UserID": 1,
      "ErrMsg" : ""
    }
  ]
}
```

Error Code - Person Failure Error Codes

Code	Description
1	Person information parameter exception, see details for specific reason
2	Person photo URL download failed
3	Person photo image too large, must be less than 500KB
4	Face feature code download failed
5	Fingerprint feature code download failed

Code	Description
6	Palm vein feature code download failed
20	Duplicate face feature code
21	Duplicate fingerprint feature code
22	Duplicate palm vein feature code
23	Duplicate card number
24	Person storage full
25	Error occurred while querying data
26	Error occurred while saving person photo
27	Face feature code format error
28	Fingerprint feature code format error
29	Palm vein feature code format error
30	Photo not recognizable
31	Error occurred while saving fingerprint feature code
32	Error occurred while saving palm vein feature code

Return Value Parameter Description

Parameter Name	Type	Required	Description
Success	int	Yes	1--indicates success Others are error codes, need to indicate error content in Message; 0--indicates failure
Message	string	No	Error information description, see Chapter 3 for details

Return Value Example

```
{
  "Success":1
}
```

 image-20210616205111411

API - Pull People to Delete

Brief Description:

- Called after sending heartbeat packet when server returns that there are people to delete. Loop call.
- Stop loop conditions: 1. Success:1, and deleteInfo is empty array or this field is absent; 2. Success:0

Request URL:

- http://Server_IP:port/People/DeletePeopleList

Request Method:

- POST

Content-Type

- application/json

Request Parameters:

Field	Type	Length	Required	Description
SN	string	30	Yes	Device ID
Limit	int	10	Yes	Limit on number of people returned per request (default 50 if not specified, max 1000)

Request Parameter Example

```
{
  "SN": "FC-8200H12345678"
}
```

Return Value Parameter Description

Parameter Name	Type	Required	Description
Success	int	Yes	1--indicates success Others are error codes, need to indicate error content in Message; 0--indicates failure
Message	string	No	Error information description, see Chapter 3 for details
DeleteAll	int	No	1: clear all person information; 0: delete by specified user ID
DeleteCount	int	Yes	Number of people to delete
DeleteList	[]	No	When DeleteAll=0, this parameter contains user IDs to delete

Return Value Examples

```
{
  "Success":1,
  "DeleteAll":0,
  "DeleteList":[
    1,2,3,4,5
  ]
}
```

API - Feedback Delete People Operation Result

Brief Description:

- After pulling a batch of people from the server and importing them, call this interface to return the import result.

Request URL:

- http://Server_IP:port/People/DeletePeopleListResult

Request Method:

- POST

Content-Type

- application/json

Request Parameters:

Field	Type	Length	Required	Description
SN	string	30	Yes	Device ID
DeleteCount	int	20	Yes	Number of deletions
DeleteAll	int	20	Yes	1: deleted all people; 0: only deleted specified people
DeleteList	[]		Yes	Deleted person IDs

Request Parameter Example

```
{
  "SN":"FC-8200H12345678",
  "DeleteCount":5,
  "DeleteAll":0,
  "DeleteList":[1,2,3,4,5]
}
```

Return Value Parameter Description

Parameter Name	Type	Required	Description
Success	int	Yes	1--indicates success Others are error codes, need to indicate error content in Message; 0--indicates failure
Message	string	No	Error information description, see Chapter 3 for details

Return Value Example

```
{
  "Success": 1
}
```

API - Push Person Information

Brief Description:

- When a person is added or modified on the device, upload the changed person information to the cloud platform.
- When the server requests to upload a specific person, push the person information to the platform.
- When the server requests to upload all people, push the person information to the platform.

Request URL:

- http://Server_IP:port/People/PushPeople

Request Method:

- POST

Content-Type

- multipart/form-data

Request Parameters:

Field	Type	Length	Required	Description
SN	string	30	Yes	Device ID
PushType	int	1	Yes	Person change type on device: 1--add; 2--update; 3--delete; 4--query;
Detail	class		No	Person details, absent if person does not exist
Photo	file		No	Upload photo file if person has a photo

Field	Type	Length	Required	Description
UserID	uint32		Yes	Person ID, the ID of the person being pushed this time

Request Parameter Example - with person photo, fingerprint feature code, palm vein feature code

```

POST /note/insertNoteFace HTTP/1.1
Accept: */*
Host: localhost:5000
Accept-Encoding: gzip, deflate, br
Connection: keep-alive
Content-Type: multipart/form-data; boundary=-----
-506873351428002157394455
Content-Length: 17842

-----506873351428002157394455
Content-Disposition: form-data; name="SN"

FC-8200H12345678
-----506873351428002157394455
Content-Disposition: form-data; name="PushType"

1
-----506873351428002157394455
Content-Disposition: form-data; name="Detail"
Content-Encoding: gzip
//The following content needs to be compressed via gzip before transmission. JSON
formatting website: https://www.json.cn/
{
  "UserID": "3",
  "Name": "888888",
  "Job": "Development",
  "Department": "Sales Department",
  "IdentityCard": "",
  "Attachment": "",
  "Photo": "http://abc.com/photo/1.jpg",
  "PhotoMD5": "613D870CA99EDF074BEE4387BAB09070",
  "PhotoLen": 55020,
  "Password": "2222",
  "CardNum": "6666",
  "AccessType": 0,
  "ExpirationDate": 0,
  "OpenTimes": 65535,
  "KeepOpen": 1,
  "Timegroup": 6,
  "Holidays": "1,3,9,10,11,17,21,25,27,30",
  "Elevators": "1,2,3,4,5",
  "FaceFeature": "abcdefgrtygnfhgfjkh ... jhkgjhkgj==t",
  "FaceFeatureMD5": "613D870CA99EDF074BEE4387BAB09070",
  "Fingerprints": [
    {

```

```

        Num: 1,
        Data: "abcdefgrtygnfhgfhjk ... jhkghjkgj==",
        MD5: "abcdefg"
    }
],
"Palmveins": [
    {
        Num: 1,
        Data: "abcdefgrtygnfhgfhjk ... jhkghjkgj==",
        MD5: "abcdefg"
    }
]
}

-----506873351428002157394455
Content-Disposition: form-data; name="Photo"; filename="Photo.jpg"
Content-Type: image/jpeg

*****JPEG file binary content*****
-----506873351428002157394455--

```

Return Value Parameter Description

Parameter Name	Type	Required	Description
Success	int	Yes	1--indicates success Others are error codes, need to indicate error content in Message; 0--indicates failure
Message	string	No	Error information description, see Chapter 3 for details

Return Value Example

```

{
  "Success": 1
}

```

Online Authentication

API - Request Server Authentication

Brief Description:

- Use this interface when there are new attendance records on the device.
- This interface only supports uploading one attendance record at a time.

Request URL:

- <http://Server IP:port/Device/RequestAuthorization>

Request Method:

- POST

Content-Type

- multipart/form-data

Parameters:

Field	Type	Length	Required	Description
SN	string	30	Yes	Device SN
RecordDetail	string	1000	Yes	Record details, JSON string
Photo	file	50KB	No	This parameter is needed only when the record contains an image

RecordDetail Field Description

Field	Type	Length	Required	Description
RecordID	long	10	Yes	Request ID
RecordType	int	5	Yes	Event type
RecordDate	int64	20	Yes	Attendance time, Unix timestamp
UserID	string	20	No	User ID
Code	string	64	No	Person code
Name	string	64	No	Person name
IdentityCard	string	64	No	ID card
Job	string	64	No	Job title
Department	string	64	No	Department
CardNum	string	20	No	Card number
QRCode	string	128	No	QR code
IsEntry	int	1	No	Whether it's entry, 1 indicates entry, 0 indicates exit
BodyTemp	int	5	No	Measured body temperature, needs to be divided by 10
PhotoLen	int	10	No	Image file length, 0 indicates no image

RecordDetail Parameter Example

```
{
  "RecordID": 1718616771001,
  "RecordType": 3,
  "RecordDate": 1718616771,
  "UserID": "1",
  "Name": "1",
  "IdentityCard": "",
  "Job": "",
  "Department": "",
  "CardNum": "0",
  "QRCode": "",
  "IsEntry": 1,
  "BodyTemp": 0,
  "PhotoLen": 40363
}
```

Request Example

```
POST /Device/RequestAuthorization HTTP/1.1
Accept: */*
Host: localhost:5000
Accept-Encoding: gzip, deflate, br
Connection: keep-alive
Content-Type: multipart/form-data; boundary=-----
-506873351428002157394455
Content-Length: 17842
-----506873351428002157394455
Content-Disposition: form-data; name="SN"

FC-8380T12345678
-----506873351428002157394455
Content-Disposition: form-data; name="RecordDetail"
Content-Encoding: gzip
Content-Type: application/octet-stream

//When request compression is enabled, the following content needs to be
compressed via gzip before transmission.
{ "RecordID":120,"RecordType":3,"RecordDate":1718616771,
  "UserID":"1","Name":"1","IdentityCard":"","Job":"","Department":"","CardNum":"0"
  ,"QRCode":"","IsEntry":1,"BodyTemp":0,"PhotoLen":40363}
-----506873351428002157394455
Content-Disposition: form-data; name="pic"; filename="Postman_file.jpg"
Content-Type: image/jpeg

*****JPEG file binary content*****
-----506873351428002157394455--
```

Return Example

```
{
  "Success": 1
}
```

Request Content Compression

- If the recordJson field in formdata is relatively long, you can use the content compression option, carrying Content-Encoding: gzip in the section.

```
Content-Disposition: form-data; name="RecordDetail"
Content-Encoding: gzip

****Compressed binary content****
```

Return Parameter Description

Parameter Name	Type	Required	Description
Success	int	Yes	See Success Return Results table for details
Message	string	No	When door opening is prohibited, the server explains the reason for prohibition And the device displays it on screen for the person checking in.

Success Return Results

Value	Description
0	Authentication failed, door opening denied
1	Authentication successful, open door and display authentication message
2	Authentication successful, open door, do not display message

Record Management

API - Upload Attendance Record

Brief Description:

- Use this interface when there are new attendance records on the device.
- This interface only supports uploading one attendance record at a time.

Request URL:

- <http://Server IP:port/Record/UploadIdentifyRecord>

Request Method:

- POST

Content-Type

- multipart/form-data

Parameters:

Field	Type	Length	Required	Description
SN	string	30	Yes	Device SN
RecordDetail	string	1000	Yes	Record details, JSON string
Photo	file	50KB	No	This parameter is needed only when the record contains an image

RecordDetail Field Description

Field	Type	Length	Required	Description
RecordID	long	10	Yes	Record sequence number
RecordType	int	5	Yes	Event type
RecordDate	int64	20	Yes	Attendance time, Unix timestamp
UserID	string	20	No	User ID
Code	string	64	No	Person code
Name	string	64	No	Person name
IdentityCard	string	64	No	ID card
Job	string	64	No	Job title
Department	string	64	No	Department
CardNum	string	20	No	Card number
QRCode	string	128	No	QR code
IsEntry	int	1	No	Whether it's entry, 1 indicates entry, 0 indicates exit
BodyTemp	int	5	No	Measured body temperature, needs to be divided by 10
PhotoLen	int	10	No	Image file length, 0 indicates no image

RecordType Event Types

Value	Description
1	Card verification
2	Fingerprint verification
3	Face verification
4	Card + Fingerprint
5	Face + Fingerprint
6	Card + Face
7	Card + Password
8	Face + Password
9	Fingerprint + Password
10	Password verification (User ID + Password)
11	Card + Fingerprint + Password
12	Card + Face + Password
13	Fingerprint + Face + Password
14	Card + Fingerprint + Face
15	Duplicate verification
16	Validity expired
17	Access time period expired
18	Cannot open door during holidays
19	Unregistered user
20	Detection lockout
21	Valid access count exhausted
22	Verification during lockout, door opening prohibited
23	Lost card
24	Blacklist card
25	Verification-free door opening -- user ID is 0 for fingerprint, card number for card verification
26	Card verification prohibited -- when card is disabled in [Permission Authentication Mode]

Value	Description
27	Fingerprint verification prohibited -- when fingerprint is disabled in [Permission Authentication Mode]
28	Controller expired
29	Verification passed -- validity about to expire
30	Abnormal body temperature, entry denied
31	Visitor password door opening
32	Dynamic QR code scan door opening
33	Person added in device menu
34	Person modified in device menu
35	Person deleted in device menu
36	Palm vein recognition
37	Card + Palm Vein + Face
38	Palm Vein + Password
39	Card + Palm Vein
40	Face + Palm Vein
41	Card + Palm Vein + Password
42	Palm Vein + Face + Password
43	Fingerprint + Palm Vein + Face
44	Combined verification -- waiting for next person
45	Combined verification failed
46	Combined verification completed
47	ID card vs. face comparison
48	Card not registered
49	QR code not registered

RecordDetail Parameter Example

```
{
  "RecordID": 120,
  "RecordType": 3,
  "RecordDate": 1718616771,
  "UserID": "1",
  "Name": "1",
```

```
"IdentityCard": "",
"Job": "",
"Department": "",
"CardNum": "0",
"QRCode": "",
"IsEntry": 1,
"BodyTemp": 0,
"PhotoLen": 40363
}
```

Request Example

```
POST /note/insertNoteFace HTTP/1.1
Accept: */*
Host: localhost:5000
Accept-Encoding: gzip, deflate, br
Connection: keep-alive
Content-Type: multipart/form-data; boundary=-----
-506873351428002157394455
Content-Length: 17842
-----506873351428002157394455
Content-Disposition: form-data; name="SN"

FC-8380T12345678
-----506873351428002157394455
Content-Disposition: form-data; name="recordJson"
Content-Encoding: gzip
Content-Type: application/octet-stream

//When request compression is enabled, the following content needs to be
compressed via gzip before transmission.
{ "RecordID":120,"RecordType":3,"RecordDate":1718616771,
"UserID":"1","Name":"1","IdentityCard":"","Job":"","Department":"","CardNum":"0"
,"QRCode":"","IsEntry":1,"BodyTemp":0,"PhotoLen":40363}
-----506873351428002157394455
Content-Disposition: form-data; name="pic"; filename="Postman_file.jpg"
Content-Type: image/jpeg

*****JPEG file binary content*****
-----506873351428002157394455--
```

Return Example

```
{
  "Success": 1
}
```

Return Error Codes

```
{
  "Success":401 //Returning 401 indicates the device is not authorized. The
device will no longer send attendance records or device parameters to the server,
but will still periodically send keep-alive packets.
}
//Device returns 401 indicating this record was not uploaded successfully
```

- Device waits up to 30 seconds for server response;
- If the server returns an error code, wait 10 seconds before re-uploading the record;

Request Content Compression

- If the recordJson field in formdata is relatively long, you can use the content compression option, carrying Content-Encoding: gzip in the section.

```
Content-Disposition: form-data; name="recordJson"
Content-Encoding: gzip

****Compressed binary content****
```

Return Parameter Description

Parameter Name	Type	Required	Description
Success	int	Yes	1--indicates success Others are error codes, need to indicate error content in Message; 0--indicates failure

API - Upload System Records

Brief Description:

- Use this interface when there are new system records on the device.
- This interface uploads system records in batches.

Request URL:

- <http://Server IP:port/Record/UploadSystemRecord>

Request Method:

- POST

Content-Type

- application/json

Parameters:

Field	Type	Length	Required	Description
SN	string	30	Yes	Device SN
RecordType	int	1	Yes	Record type
Records	[]	1000	Yes	Record list

Record Field Description

Field	Type	Length	Required	Description
RecordID	long	10	Yes	Record sequence number
RecordType	int	5	Yes	Event type 1-door sensor record; 2-system record
RecordDate	int64	20	Yes	Time, Unix timestamp

RecordType Field Description

Value	Description
1	Door sensor record
2	System record

RecordType -- Door Sensor Record Event Types

Value	Description
1	Door sensor - door open
2	Door sensor - door closed
3	Entered door sensor alarm detection state
4	Exited door sensor alarm detection state
5	Door not properly closed
6	Door opened using button
7	Door locked when using button to open
8	Controller expired when using button to open

RecordType -- System Record Event Types

Value	Description
1	Software door open

Value	Description
2	Software door close
3	Software keep-open
4	Controller auto enters keep-open
5	Controller auto closes door
6	Long press exit button keep-open
7	Long press exit button normally closed
8	Software lock
9	Software unlock
10	Controller scheduled lock -- auto lock at time
11	Controller scheduled lock -- auto unlock at time
12	Alarm -- lock
13	Alarm -- unlock
14	Illegal verification alarm
15	Door sensor alarm
16	Duress alarm
17	Door open timeout alarm
18	Blacklist alarm
19	Fire alarm
20	Anti-disassembly alarm
21	Illegal verification alarm cleared
22	Door sensor alarm cleared
23	Duress alarm cleared
24	Door open timeout alarm cleared
25	Blacklist alarm cleared
26	Fire alarm cleared
27	Anti-disassembly alarm cleared
28	System powered on
29	System error reset (watchdog)
30	Device format record

Value	Description
31	Card reader reversed
32	Card reader line not properly connected
33	Unrecognized card reader
34	Network cable disconnected
35	Network cable connected
36	WIFI connected
37	WIFI disconnected
38	Bluetooth door open
39	Roll call timeout
40	All people cleared in device menu
41	People backed up to USB in device menu
42	People imported from USB in device menu
43	Indoor unit remote door open
44	All records deleted
45	All people deleted

RecordDetail Parameter Example

```
[
  {
    "RecordID": 1,
    "RecordType": 1,
    "RecordDate": 1718616771
  },
  {
    "RecordID": 2,
    "RecordType": 1,
    "RecordDate": 1718616772
  }
]
```

Parameter Example

```
{
  "SN": "FC-8380T12345678",
  "RecordType": 1,
  "Records": [
    {
```

```

        "RecordID": 1,
        "RecordType": 1,
        "RecordDate": 1718616771
    },
    {
        "RecordID": 2,
        "RecordType": 1,
        "RecordDate": 1718616772
    }
]
}

```

Return Example

```

{
  "Success": 1
}

```

Return Error Codes

```

{
  "Success":401 //Returning 401 indicates the device is not authorized. The
device will no longer send attendance records or device parameters to the server,
but will still periodically send keep-alive packets.
}
//Device returns 401 indicating this record was not uploaded successfully

```

- Device waits up to 30 seconds for server response;
- If the server returns an error code, wait 10 seconds before re-uploading the record;

Return Parameter Description

Parameter Name	Type	Required	Description
Success	int	Yes	1--indicates success Others are error codes, need to indicate error content in Message; 0--indicates failure

Tange Audio/Video Management

Cloud Intercom Audio/Video Parameters

API Description

- **API - Device Actively Pulls Working Parameters**
- Cloud intercom parameters are obtained by the device through the pull parameters interface.

Parameter Description

Parameter Name	Type	Required	Description
TangeDetail	object	Yes	Tange device SN information
BindDevice	array [object]	Yes	Bound device information list
WeChat	array [object]	Yes	Bound WeChat user information list

TangeDetail Object Field Description

Parameter Name	Type	Required	Description
SN	string	Yes	Tange device SN
Key	string	Yes	Tange device secret key
Deadline	int	Yes	Deadline, timestamp, seconds
WechatVoip	int	Yes	Whether Voip is supported 1--supported; 0--not supported
WeChatDeadline	int	Yes	WeChat Voip call deadline, timestamp, seconds

BindDevice Object Field Description

Parameter Name	Type	Required	Description
TangeSN	string	Yes	Tange device SN
SN	string	Yes	Device SN
DeviceType	int	Yes	1--outdoor unit; 2--indoor unit
Name	string	Yes	Device name, used for display in list
BuildCode	string	Yes	Building number, numeric string, max 8 characters

Parameter Name	Type	Required	Description
RoomCode	string	Yes	Room number, numeric string, max 8 characters

WeChat Object Field Description

Parameter Name	Type	Required	Description
Name	string	Yes	Username
WeChatId	string	Yes	WeChat OpenId
BuildCode	string	No	Building number, numeric string, max 8 characters
RoomCode	string	No	Room number, numeric string, max 8 characters

Return Example

```
{
  "TangeDetail": {
    "SN": "TGxxxx",
    "Key": "*****",
    "Deadline": 123456,
    "wechatVoip": 1,
    "wechatDeadline": 123456
  },
  "BindDevice": [
    {
      "SN": "FC-xxxx",
      "TangeSN": "TGxxxx",
      "BuildCode": "8888",
      "RoomCode": "0001"
    },
    {
      "SN": "FC-xxxx",
      "TangeSN": "TGxxxx",
      "BuildCode": "8888",
      "RoomCode": "0002"
    }
  ],
  "wechat": [
    {
      "Name": "Zhang San",
      "wechatId": "abcdefg.....",
      "BuildCode": "8888",
      "RoomCode": "1004"
    }
  ]
}
```

```
    },
    {
      "Name": "Li Si",
      "WeChatId": "abcdefg.....",
      "BuildCode": "8888",
      "RoomCode": "1905"
    },
  ],
}
```

API - Query Device Connection Token

Brief Description:

- When the device needs to use the Tange video intercom platform to cloud-call corresponding devices, this interface is used to obtain the Token needed by the Tange SDK to connect to the device.

Request URL:

- <http://Server IP:port/Device/Tangelot/ConnectDevice>

Request Method:

- POST

Content-Type

- application/json

Request Parameters

Field	Type	Length	Required	Description
SN	string	30	Yes	Device ID
ConnectList	array		Yes	Tange device SNs to connect

Request Parameter Example

```
{
  "SN": "FC-8200H12345678",
  "ConnectList": [
    "TGxxxxx1", //Tange device SN
    "TGxxxxx2",
    "TGxxxxx3"
  ]
}
```

Return Value Parameter Description

Parameter Name	Type	Required	Description
Success	int	Yes	1--indicates success Others are error codes, need to indicate error content in Message; 0--indicates failure
Message	string	No	Error information description, see Chapter 3 for details
Tokens	array	Yes	Tokens obtained for connecting to devices

Tokens Object Field Description

Parameter Name	Type	Required	Description
SN	string	Yes	Device SN
Token	string	Yes	Connection Token

Return Example

```
{
  "Success": 1,
  "Tokens": [
    {
      "SN": "TGxxxx",
      "Token": "*****"
    },
    {
      "SN": "TGxxxx",
      "Token": "*****"
    }
  ]
}
```